Assignment_01

MAT216_Sec_01_02_03

Summer 2024

Remarks: You will have to submit your assignment on 26 June 2024 (Wednesday, Class time). No late submission will be accepted. Total marks is 40. It will be converted to 20.

1. (i) Given $A = \begin{bmatrix} -1 & 0 & -3 \\ -2 & 2 & -2 \\ 5 & -4 & 7 \end{bmatrix}$. Find λ such that $\det(A - \lambda I_3) = 0$. [7]

   (ii) Write the general solution in vector form of the linear system represented by the augmented matrix. [3]

$$[A|b] = \begin{bmatrix} 1 & -7 & 2 & -5 & 8 \\ 0 & 1 & -3 & 3 & 1 \\ 0 & 0 & 0 & 1 & -1 \end{bmatrix}$$

2. Mr. X and Mr. Y rent their ice out to the community whenever possible. Mr. X charges a flat rate of $20.00 plus $15.00 for every hour rented. Mr. Y charges $50.00 for a flat rate but only asks for $10.00 for every hour rented. [10]

   (i) Write an equation to model the cost of renting the ice surface for each arena?
   (ii) Determine the intersection point of the equations. What does this intersection point represent?
   (iii) Explain when it is best to use Mr. X's service and when it is best to use Mr. Y's service.

3. Solve the following system by using Gauss-Jordan elimination: [10]

$$\begin{gathered} 5x_3 + 15x_5 = 5 \\ 2x_2 + 4x_3 + 7x_4 + x_5 = 3 \\ x_2 + 2x_3 + 3x_4 = 1 \\ x_2 + 2x_3 + 4x_4 + x_5 = 2 \end{gathered}$$

4. (i) **Definition**: An orthogonal matrix is a square matrix A if and only if its transpose is as same as its inverse. i.e., $A^T = A^{-1}$, where A^T is the transpose of A and A^{-1} is the inverse of A. [7]

Problem: If $A = \begin{bmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}$ is an orthogonal matrix, find $(AA^T)^{-1}$.

(ii) Solve the following system of equation using the concept of multiplying with the inverse of the coefficient matrix (A^{-1}) in the matrix equation $Ax = b$. [3]

$$2x + 3y = -1, \;\; 3x + 2y = 1$$